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Printed Page: 1 of 2s

Subject Code:BP303T

Roll No:

BPHARM
(SEM. III) THEORY EXAMINATION 2025-26
PHARMACEUTICAL MICROBIOLOGY

TIME: 3 HRS**M.MARKS: 75**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10 x 2 = 20

a.	Define prokaryotic cell.
b.	What is IMViC test?
c.	Define sterilization indicator with one example.
d.	What is acid-fast staining?
e.	Define virus.
f.	What is phenol coefficient?
g.	Define clean area classification.
h.	What is microbiological assay?
i.	Define viable count.
j.	What is an established cell culture?

SECTION B

2. Attempt any *two* parts of the following:

2 x 10 = 20

a.	Describe the principle, procedure, applications, merits and demerits of radiation sterilization.
b.	Explain the morphology, classification, reproduction and cultivation of fungi.
c.	Discuss the sterility testing of sterile pharmaceutical products as per IP, BP and USP.

SECTION C

3. Attempt any *five* parts of the following:

7 x 5 = 35

a.	Differentiate between prokaryotic and eukaryotic cells with suitable examples.
b.	Describe quantitative measurement of bacterial growth by total count and viable count methods.
c.	Explain the classification and mode of action of disinfectants. Discuss factors affecting disinfection.
d.	Explain the design of aseptic areas and discuss various sources of contamination and their prevention.
e.	Describe the principles and methods of microbiological assay. Add a note on standardization of antibiotics.
f.	Explain the preservation of pharmaceutical products using antimicrobial agents and discuss evaluation of microbial stability.
g.	Discuss primary, established and transformed cell cultures along with their applications in pharmaceutical industry and research.